MoleHole: Real-time Black Hole Simulation

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Abstract

bla bla bla

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Todo list

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Part I

Introduction

1 Introduction

This is a technical research paper covering the scientific part of the development of the open-source MoleHole project[1]. It is not a user manual, neither a tutorial, and it is not particularly intended to be read linearly. This paper aims to cover both the very foundational concepts of computer graphics, physics and mathematics, as well as highly specific technical details about general relativistic ray tracing and physical simulations. The reader is encouraged to read the table of contents first and jump between sections of interest.

1.1 Motivation

Inspired by the visual display of the black hole *Gargantua* in the movie *Interstellar*[15], and a general interest in computer graphics and physics, we decided to build a black hole simulator. What started as a simple school project soon evolved into a more ambitious project, aiming to create an interactive application able to export physically accurate renderings of black holes and other stellar objects.

For *Interstellar*, the visual effects company "Double Negative" developed the Double Negative Gravitational Renderer (DNGR), which traces bundles of light rays through the spacetime of a spinning black hole instead individual ones[8]. The spin was lowered from $a/M = 0.999$ to $a/M = 0.6$ because the flattened shadow edge and multiple disk images were deemed too confusing for a general audience[8]. The same code was later used for scientific publications. For example the visualization of the movie's wormhole[9].

In 2019 the Event Horizon Telescope (EHT), a global network of radio telescopes combined through Very Long Baseline Interferometry (VLBI) at a wavelength of 1.3 mm, published the first image of the supermassive black hole M87*[4]. In 2022 the image of Sagittarius A*, the black hole at the center of the Milky Way Galaxy[5]. Figure 1.1 shows these images next to the DNGR rendering.

1.2 Related Work

Planned: DNGR, GYOTO, GRay, RAPTOR, geokerr, GeodesicViewer, Riazuelo.

1.3 Problem Statement and Contributions

Planned: what is new, tiered renderer, spectral g-shifted transfer, Kerr–Schild orientation.

1.4 Accuracy Ladder and Scope

Planned: L0 exact, L1 modelled, L2 approximate, L3 not modelled.

1.5 Outline and Reading Paths

Planned: prerequisites, rendering reader vs. physics reader.

1.6 Notation and Conventions

Planned: signature $(-+++)$, $G = c = 1$, index conventions.

[1] `https://github.com/g-martin772/MoleHole`



(a) *Interstellar* (DNGR), $a/M = 0.6$ **(b)** M87*, EHT 2017 **(c)** Sgr A*, EHT 2022

Figure 1.1: A simulated and two observed black hole images. (a) Rendered with DNGR. A variant of the disk seen in *Interstellar*[8]. (b) M87*[4] (c) Sagittarius A*[5]

Part II

Mathematical Foundations

2 Space and Coordinates

Scope: From points and vectors to the celestial sphere.

2.1 Points, Vectors, Frames

Planned: affine vs. vector space, handedness, math vs. Vulkan convention.

2.2 Cartesian Coordinates

Planned: orthonormal basis, distance, dot and cross product.

2.3 Curvilinear Coordinates

Planned: polar, cylindrical, spherical, physics vs. math (θ, ϕ), charts, Jacobian, scale factors.

2.4 Coordinate and Orthonormal Bases

Planned: components change, vectors do not.

2.5 Calculus in Curvilinear Coordinates

Planned: line, area, volume elements, ∇, solid angle $\mathrm{d}\Omega = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$.

2.6 Oblate Spheroidal Coordinates

Planned: Kerr's natural coordinates, relation to Boyer–Lindquist.

2.7 Coordinate Singularities

Planned: poles, axis, gimbal lock, pinch artefacts.

2.8 Rotations and Orientation

Planned: $SO(3)$, axis–angle, Euler, quaternions, slerp, aligning a spin axis to $+z$.

2.9 Homogeneous Coordinates and Projection

Planned: affine maps, pinhole, NDC, camera-relative origin.

2.10 The Celestial Sphere (opt.)

Planned: equirectangular, cubemap, sampling distortion, HEALPix.

3 Calculus, Variational Principles, Linear Algebra

Scope: The analysis that geodesics and Hamiltonians rest on.

3.1 Taylor Series and Asymptotics

Planned: big-O, order of approximation, weak-field expansions.

3.2 Multivariable Calculus

Planned: chain rule, Jacobian, Hessian, change of variables, implicit function theorem.

3.3 Singular Integrals

Planned: turning-point $1/\sqrt{\cdot}$ singularity, substitutions.

3.4 Ordinary Differential Equations

Planned: IVP, Picard–Lindelöf, autonomous systems, first integrals, phase space, stability.

3.5 Calculus of Variations

Planned: Euler–Lagrange, Fermat's principle, Lagrangian mechanics, Noether and conserved quantities.

3.6 Hamiltonian Mechanics

Planned: Legendre transform, Hamilton's equations, Poisson bracket, Liouville theorem, Hamilton–Jacobi.

3.7 Linear Algebra

Planned: inner products, eigen/SVD, Gram–Schmidt, indefinite inner products, 4×4 inverse and determinant.

3.8 Special Functions and Polynomial Roots (opt.)

Planned: elliptic integrals (Legendre, Carlson), Jacobi functions, cubic/quartic roots for ISCO and radial potential.

4 Differential Geometry and Tensor Calculus

Scope: Everything needed to read the geodesic equation.

4.1 Intuition: Intrinsic vs. Extrinsic Curvature

Planned: the 2-sphere as running example.

4.2 Manifolds and Charts

Planned: atlases, coordinate transformations, why the sphere needs two charts.

4.3 Tangent Vectors

Planned: curves, directional derivatives, coordinate basis ∂_μ.

4.4 Covectors and One-Forms

Planned: dual basis, contra- vs. covariant.

4.5 Tensors

Planned: transformation law, products, contraction, symmetry, Einstein summation.

4.6 The Metric

Planned: line element, signature, raising/lowering, $\sqrt{|g|}$, null/time/space-like, causal structure.

4.7 Connection and Covariant Derivative

Planned: why ∂ fails, Christoffel symbols, metric compatibility.

4.8 Parallel Transport and Geodesics

Planned: variational and autoparallel derivations, affine parameter, null geodesics.

4.9 Curvature

Planned: Riemann, geodesic deviation, Ricci, Einstein, Weyl, Kretschmann, coordinate vs. true singularity, Bianchi.

4.10 Killing Vectors and Conserved Quantities

Planned: Killing vectors and tensors, Lie derivative, conserved quantities along geodesics.

4.11 Frame Fields and Tetrads

Planned: orthonormal frames, local Lorentz frame, spin connection.

4.12 Hamiltonian Formulation on a Manifold

Planned: $H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu$, covariant momenta, symplectic structure.

4.13 Differential Forms and Advanced Tools (opt.)

Planned: exterior derivative, Cartan structure equations, Newman–Penrose, Petrov types.

4.14 Worked Examples

Planned: flat space in spherical coordinates ($\Gamma \neq 0$ but $R = 0$), 2-sphere, hyperbolic plane, embedding diagrams.

5 Numerical Computation

Scope: What the machine does to our equations.

5.1 Integers and Overflow

Discrete quantities are represented as binary signed or unsigned integers, which can overflow when their values exceed the maximum or minimum representable values. The range of an n-bit signed integer is $[-2^{n-1}, 2^{n-1} - 1]$, while an unsigned integer has a range of $[0, 2^n - 1]$.

When an integer exceeds its maximum value, it wraps around to the minimum value (and vice versa). Therefore, it is crucial to use sufficiently large integer types and/or implement overflow checking mechanisms. This requires careful consideration of expected number ranges.

5.2 Floating-Point Representation and Numerical Errors

Computers approximate real numbers using finite-precision representations governed by the IEEE 754 standard for floating-point arithmetic[7]. Compared to integer representations, floating-point numbers can represent a much wider range of values, by using a normalized scientific notation with a fixed number of significant digits (the mantissa) and an exponent.

Floating-point systems introduce fundamental error sources[6]:

- **Rounding and Machine Epsilon:** Real numbers cannot be represented with infinite precision. The spacing between adjacent representable floating-point numbers is bounded by the Machine Epsilon ϵ_{mach}. Operations on physical quantities introduce rounding errors that accumulate over multistep time integrations (e.g., Verlet or Runge-Kutta integrators). While single precision (32-bit) is the de facto standard for most programming languages, it is insufficient for large-scale simulations with both high number ranges and small deltas. Double precision (64-bit) and quadruple precision (128-bit) are also available on modern processors, but come with increased memory usage and computational cost. See Table 5.1 for a comparison of IEEE 754 formats.
- **Catastrophic Cancellation:** Subtracting two nearly equal floating-point numbers results in loss of significant digits. This frequently occurs when computing relative position vectors between close particles ($\vec{r}_1 - \vec{r}_2$), leading to increased relative errors.
- **Underflow, Overflow, and NaN:** Exceeding the dynamic range of the exponent field results in overflow (yielding $\pm\infty$) or underflow (yielding subnormal numbers or zero), which can corrupt calculations. A floating-point values might also hold the special value NaN (Not a Number) to indicate undefined or unrepresentable results, such as $0/0$ or $\infty - \infty$.

5.3 GPU Floating-Point Specifics

Planned: fma, fast-math, trig and rsqrt accuracy, fp64 throughput on consumer GPUs.

5.4 Error Analysis

Planned: condition number, forward/backward error, error growth, compensated summation, double-double.

5.5 Nondimensionalization and Camera-Relative Coordinates

Planned: r/M scaling as the f32 strategy.

5.6 Automatic Differentiation

Planned: forward-mode dual numbers vs. finite differences vs. symbolic, reverse mode.

5.7 Root Finding and Interpolation

Planned: bisection, Newton, Brent, LUTs, Chebyshev interpolation.

Table 5.1: Comparison of IEEE 754 floating-point formats[7]

Format	**Digits**	**($\epsilon_{\mathbf{mach}}$)**	**Dynamic Range**	**Relative Error ($50^{\mathbf{th}}$ / $99^{\mathbf{th}}$)**
`f32`	$6-7$	$\approx 1.19 \times 10^{-7}$	1.18×10^{-38} to 3.40×10^{38}	$\sim 10^{-7}$ / $\sim 10^{-5}$
`f64`	$15-17$	$\approx 2.22 \times 10^{-16}$	2.23×10^{-308} to 1.79×10^{308}	$\sim 10^{-16}$ / $\sim 10^{-14}$
`f128`	34	$\approx 1.92 \times 10^{-34}$	3.36×10^{-4932} to 1.18×10^{4932}	$\sim 10^{-34}$ / $\sim 10^{-32}$

6 Numerical Integration of ODEs

Scope: The geodesic solver toolbox.

6.1 Initial Value Problems

Planned: local/global error, order, consistency, stability, convergence.

6.2 Elementary One-Step Methods

Planned: Euler, midpoint, Heun.

6.3 Runge–Kutta Methods

Planned: Butcher tableau, order conditions, RK4 and its error.

6.4 Embedded Pairs and Adaptive Step Control

Planned: RKF45, Cash–Karp, Dormand–Prince 5(4) FSAL, I/PI controllers, tolerances, step rejection.

6.5 Dense Output (opt.)

Planned: continuous extension, intersecting geometry between steps.

6.6 Stability and Stiffness (opt.)

Planned: stability regions, implicit methods, Gauss–Legendre.

6.7 Extrapolation and Multistep Methods (opt.)

Planned: Richardson, Bulirsch–Stoer, Adams.

6.8 Geometric Integration

Planned: symplectic maps, Verlet/leapfrog, Yoshida and Forest–Ruth, backward error analysis, bounded vs. drifting energy.

6.9 Non-Separable Hamiltonians

Planned: implicit midpoint, extended phase space (Pihajoki, Tao), explicit symplectic Kerr integrators, when RK beats symplectic for null rays.

6.10 Constraints, Invariants, Time Transformations

Planned: $H = 0$ drift, E, L_z, Q diagnostics, renormalization, Mino time, Preto–Tremaine.

6.11 Sensitivity and Chaos

Planned: Lyapunov exponent near the photon sphere, convergence testing, log-log order slopes, Aitken.

6.12 Event Handling

Planned: plane crossing, horizon, terminal conditions, step truncation and bisection.

6.13 Step-Size Heuristics for Rays

Planned: $\Delta\lambda = \varepsilon r$, logarithmic radial cost.

6.14 Particle Integrators

Planned: velocity Verlet, softening, spatial hashing, Barnes–Hut, determinism.

6.15 GPU Execution

Planned: one ray per thread, divergence, fixed budgets, wavefront.

7 Quadrature and Monte Carlo

Scope: Deterministic and stochastic integration.

7.1 Quadrature

Planned: trapezoid, Simpson, Gauss–Legendre, adaptive, singular substitution, worked example: the deflection integral.

7.2 Probability Recap

Planned: pdf/CDF, expectation, variance, law of large numbers, CLT.

7.3 Monte Carlo Integration

Planned: estimator, $O(N^{-1/2})$, bias vs. variance.

7.4 Variance Reduction

Planned: importance, stratified, control variates, Russian roulette, MIS.

7.5 Sampling Techniques

Planned: inverse CDF, rejection, sphere/hemisphere/cosine, sampling a spectrum or blackbody.

7.6 Low-Discrepancy Sequences (opt.)

Planned: Halton, Sobol, blue noise.

7.7 Convergence and Error Bars

Planned: estimating convergence, denoising pointer.

Part III

Rendering Theory

8 Light Transport in Flat Space

Scope: Radiometry to the rendering equation.

8.1 Radiometry and Surface Scattering

This section covers the fundamental concepts of radiometry, photometry, and surface scattering functions that are essential for understanding light transport in rendering.

8.1.1 Radiometry and Photometry

It is important to understand the difference between radiometry (physical measurement of Electro-Magnetic (EM) radiation) and photometry (light as it is perceived by the human eye). Both systems define their own set of units. The conversion between the two is defined by the spectral luminous efficiency function $V(\lambda)$, which peaks at approximately 555 nm (green). This is why colors like green and yellow appear brighter to the human eye compared to colors like blue.

Table 8.1 shows the key radiometric quantities, and Table 8.2 lists their photometric equivalents. Note that while rendering calculations are performed using radiometric units (specifically Radiance), output devices (displays) and human perception operate in the photometric domain (Luminance). [11].

8.1.2 Bidirectional Reflectance Distribution Function (BRDF)

The Bidirectional Reflectance Distribution Function (BRDF) defines how a material scatters incoming light from direction ω_i into outgoing direction ω_o at a given surface point x [14]. Formally the BRDF is the differential ratio of reflected radiance $\mathrm{d}L_r$ to incident irradiance $\mathrm{d}E_i$:

$$f_r(x, \omega_i, \omega_o) = \frac{\mathrm{d}L_r(x, \omega_o)}{\mathrm{d}E_i(\omega_i)} \tag{8.1}$$

$$= \frac{\mathrm{d}L_r(x, \omega_o)}{L_o(y_i, -\omega_i)\,|\cos\theta_i|\,\mathrm{d}\omega_i} \tag{8.2}$$

The unit of the BRDF is radiance divided by irradiance, which is sr^{-1} (inverse steradians).

In terms of fundamental flux quantities, the BRDF can also be expressed as:

$$f_r(x, \omega_i, \omega_o) = \frac{\mathrm{d}^3\Phi_r}{\mathrm{d}^2\Phi_i\,\mathrm{d}\omega_o\,|\cos\theta_o|} \tag{8.3}$$

The BRDF must also satisfy the following requirements:

- **Positivity**: Light cannot be negative, so it must satisfy $f_r(x, \omega_i, \omega_o) \geq 0$ for all ω_i, ω_o
- **Helmholtz Reciprocity**: States that a ray and its reverse ray must follow the same optical path (e.g. reflections, refractions) $f_r(x, \omega_i, \omega_o) = f_r(x, \omega_o, \omega_i)$
- **Energy Conservation**: The cosine weighted integral of the BRDF over the hemisphere must be 1 or less for all ω_o:

$$\int_\Omega f_r(x, \omega_i, \omega_o)|\cos\theta_o|\mathrm{d}\omega_o \leq 1$$

8.1.3 Bidirectional Transmittance Distribution Function (BTDF)

While the BRDF covers reflection, the Bidirectional Transmittance Distribution Function (BTDF) handles light that passes through a surface (transmission). The lower hemisphere Ω^- is used for incoming light (BRDF), and the upper hemisphere Ω^+ is used for outgoing light (BTDF). Unlike reflection, the BTDF is not symmetric, as light can be transmitted in one direction but not the other, and when light transmits across an interface between two media, with different refractive indices n_1 and n_2, the BTDF is scaled by the ratio of the squared indices of refraction $(n_2/n_1)^2$ according to Snell's law. [2, 14]

$$f_t(\omega_i, \omega_t) = \frac{dL_t(\omega_t)}{dE_i(\omega_i)} = \frac{dL_t(\omega_t)}{L_i(\omega_i)|\cos\theta_i|d\omega_i}$$

8.1.4 Bidirectional Scattering Distribution Function (BSDF)

The Bidirectional Scattering Distribution Function (BSDF) unifies reflection and transmission into a single function over the entire sphere of directions 4π steradians, logically representing the sum of f_r and f_t. [1, 16]

$$f_s(\omega_i, \omega_o) = f_r(\omega_i, \omega_o) + f_t(\omega_i, \omega_t)$$

Depending on whether the incident and outgoing directions are on the same side of the surface (reflection) or opposite sides (transmission), the BSDF decomposes into two distinct components, because light cannot simultaneously reflect and transmit along the exact same outgoing vector ω_o at a sharp boundary point. Therefore, the general BSDF is defined piecewise as:

$$f_s(x, \omega_i, \omega_o) = \begin{cases} f_r(x, \omega_i, \omega_o), & \text{reflection} \\ f_t(x, \omega_i, \omega_o), & \text{transmission} \end{cases} \tag{8.4}$$

8.1.5 Isotropic vs Anisotropic Scattering

Scattering at a point depends on 4 angular dimensions: the incident direction (θ_i, ϕ_i) and the outgoing direction (θ_o, ϕ_o). [16, 17]

Table 8.1: Radiometric Quantities (Physical) [11]

Quantity	Symbol	Unit	Description
Radiant Energy	Q_e	Joule [J]	Total energy of radiation
Radiant Flux (Power)	Φ_e	Watt [W]	Energy per unit time (J/s)
Radiant Intensity	I_e	Watt/sr [W/sr]	Flux per unit solid angle
Irradiance	E_e	Watt/m^2 [W/m^2]	Incident flux per unit area
Radiance	L_e	Watt/(sr $\cdot$ m^2)	Flux per unit projected area per unit solid angle

Table 8.2: Photometric Quantities (Perceptual) [11]

Quantity	Symbol	Unit	Description
Luminous Energy	Q_v	Talbot [T]	Perceived energy of light
Luminous Flux	Φ_v	Lumen [lm]	Perceived power
Luminous Intensity	I_v	Candela [cd]	Flux per unit solid angle (lm/sr)
Illuminance	E_v	Lux [lx]	Incident flux per unit area (lm/m^2)
Luminance	L_v	Nit [cd/m^2]	Perceived brightness of a surface

Isotropic Scattering: A surface is isotropic if its scattering properties are invariant under rotation about the surface normal n. Spinning the object around its normal does not change the appearance of the surface. Therefore the scattering function no longer depends on the azimuthal angles ϕ_i and ϕ_o, and the BSDF reduces to a function of only the polar angles θ_i and θ_o: $f_s(\theta_i, \theta_o, \Delta\phi)$ Examples of isotropic materials include Lambertian surfaces, metals, dielectrics, glass, and some plastics.

Anisotropic Scattering: Anisotropic surfaces have scattering properties that vary with rotation about the surface normal. This leaves the BSDF as a 4D function of both the polar and azimuthal angles: $f_s(\theta_i, \phi_i, \theta_o, \phi_o)$. Anisotropy commonly results from an oriented microscopic structure. For example mie scattering from aligned fibers, water droplets, brushed metals, or dust.

In practice, anisotropic models often use a tangent direction t in addition to the surface normal n to describe this orientation. The tangent and bitangent define a local frame in which the microfacets or scattering lobes can have different roughness values along different axes. When the roughness is equal in both tangent directions, the model reduces to the isotropic case.

8.2 Microfacet Models and Fresnel

Planned: GGX, Smith masking-shadowing, Fresnel, energy conservation, multiple scattering, BSSRDF.

8.3 The Rendering Equation

The fundamental mathematical model for simulating light in physically-based rendering is the Rendering Equation, introduced by Kajiya in 1986[10]. It describes the equilibrium distribution of radiance in a scene.

8.3.1 The Equation

The Rendering Equation expresses the total spectral radiance L_o leaving a point x in direction ω_o as the sum of emitted radiance L_e and reflected radiance:

$$L_o(x, \omega_o) = L_e(x, \omega_o) + \int_\Omega f_r(x, \omega_i, \omega_o) L_i(x, \omega_i)(\omega_i \cdot n) d\omega_i \tag{8.5}$$

Where:

- $L_o(x, \omega_o)$ is the total outgoing radiance.
- $L_e(x, \omega_o)$ is the emitted radiance (for light sources).
- Ω is the unit hemisphere centered at normal n containing all incoming directions ω_i.
- $f_r(x, \omega_i, \omega_o)$ is the BRDF, defining how light is scattered.
- $L_i(x, \omega_i)$ is the incoming radiance from direction ω_i.
- $(\omega_i \cdot n)$ is the geometry term (Lambert's cosine law), accounting for the spreading of light over the surface area (often written as $\cos\theta_i$).

Usage of Radiance (L) Radiance (L) is the central quantity in rendering for several reasons:

1. **Invariance**: Radiance is constant along a ray in a vacuum. This implies $L_i(x, \omega) = L_o(y, -\omega)$ where y is the visible surface point. This property enables ray tracing: to find the radiance reaching the camera, we simply trace a ray into the scene and calculate the radiance leaving the intersected surface.
2. **Sensor Response**: Camera sensors and the human eye respond to radiance (integrated over the pixel area and solid angle).

3. **Light Transport**: It describes the flow of light energy through space.

The recursive nature of L_i (which is the L_o from another point) makes this integral equation difficult to solve analytically. Therefore, most implementations use numerical methods like Monte Carlo integration (see Section 7.3).

8.3.2 Spherical Integration

To solve the integral numerically, we often parameterize the hemisphere Ω using spherical coordinates (θ, ϕ). The differential solid angle $d\omega_i$ changes variables as follows:

$$d\omega = \sin\theta d\theta d\phi \tag{8.6}$$

Substituting this into the rendering equation gives the explicit double integral form:

$$\int_\Omega (\dots) d\omega_i = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(x, \theta, \phi) \cdot f_r(x, \theta, \phi, \omega_o) \cos\theta \sin\theta d\theta d\phi \tag{8.7}$$

Here, θ is the polar angle (from the normal n, range $[0, \pi/2]$) and ϕ is the azimuth angle (range $[0, 2\pi]$). The term $\cos\theta$ comes from the geometry term $(\omega_i \cdot n)$, and $\sin\theta$ comes from the differential solid angle measure.

8.3.3 Color Spectrum

Radiance L varies with wavelength ($L(\lambda)$). In computer graphics, strictly accurate rendering would require integrating over the entire visible spectrum. However, for efficiency, we typically sample this spectrum at three discrete wavelengths corresponding to Red, Green, and Blue (RGB). This mimics the tristimulus response of the human eye. While sufficient for most purposes, this approximation can fail for phenomena like dispersion, fluorescence or metamerism resulting in inaccurate results.

8.3.4 Energy Conservation (White Furnace Test)

A physically-based material must conserve energy: it cannot reflect more light than it receives. The "White Furnace Test" is a thought experiment (or unit-test) used to verify this. If a non-absorbing object is placed in an environment with uniform radiance $L_i = 1$ everywhere, the outgoing radiance L_o must also be exactly 1 (as long as no other form of light is emitted). For a perfectly diffuse (Lambertian) surface with albedo ρ, the BRDF is constant $f_r = c$. Integrating the cosine term over the hemisphere yields π:

$$\int_\Omega \cos\theta d\omega = \pi$$

Therefore, to ensure $L_o = \rho L_i$, we must normalize the BRDF:

$$f_{r,\text{diffuse}} = \frac{\rho}{\pi}$$

The factor of $1/\pi$ is essential for energy conservation.

8.4 Path Formulation of Light Transport

Planned: Neumann series, path notation, measurement equation, importance.

8.5 Participating Media

Planned: absorption, emission, scattering, Beer–Lambert, optical depth, phase functions, volume rendering equation as the flat analogue of the radiative transfer chapter.

8.6 Lights and Cameras

Planned: light types and units, pinhole and thin lens, exposure (EV), sensor response.

9 Ray Tracing

Scope: From a pixel to a hit point.

9.1 Primary Rays and Pixel Filtering

Planned: ray generation, FOV, aspect, jitter, reconstruction filters.

9.2 Ray–Primitive Intersection

Planned: sphere, plane, triangle (Möller–Trumbore), SDF and sphere tracing.

9.3 Acceleration Structures

Acceleration structures are essential for reducing the computational complexity of ray-tracing from $O(N \cdot M)$ to a logarithmic scale. They utilize spatial or object subdivision to quickly cull geometry that a ray cannot intersect. This section is based on lecture 1 of a course from the TU Wien Faculty of Informatics, Institute of Visual Computing & Human-Centered Technology, Research Unit of Computer Graphics[11].

Bounding Volumes Bounding volumes simplify complex geometry into basic shapes that are computationally cheap to test against rays. Common types include:

- **Axis-Aligned Bounding Box (AABB)**: Defined by minimum and maximum extents along the coordinate axes (x, y, z). They are fast to build and test (using the slab method) but may fit underlying geometry poorly if the object is rotated, leading to more false positives during traversal.
- **Oriented Bounding Box (OBB)**: Similar to AABBs but aligned with the object's principal axes rather than the world axes. They provide a much tighter fit, reducing the number of false positive intersections, but are more expensive to build and test.
- **Bounding Sphere**: Defined by a center point and a radius. They are rotationally invariant and offer the fastest intersection test (procedural spheres work well with way math as described in Section 9.2), but often provide a bad fit for complex objects because the spheres will inevitably overlap.

Spatial vs. Object Subdivision

- **Spatial Subdivision**: The space itself is divided into cells like Grids, Octrees, or K-Dimensional Trees (K-d Trees). Objects that span boundaries must be referenced in multiple nodes, potentially increasing memory usage.
- **Object Subdivision**: The set of objects is partitioned into a hierarchy, most notably the Bounding Volume Hierarchy (BVH). A BVH is a tree structure where each node represents a bounding volume that encompasses all geometry in its children. Unlike spatial subdivision, each geometric primitive belongs to exactly one leaf node. Bounding volumes of sibling nodes can overlap, but memory consumption is generally lower and predictable ($2N - 1$ nodes for N primitives).

Building Strategies Top-down construction is a common approach, particularly for BVHs. The set of primitives is recursively split into child nodes until a leaf criterion (e.g., max triangles per node) is met. The quality of the tree depends heavily on the split strategy:

- **Spatial Median**: Splits the node at the spatial center of its bounding box. Simple but can result in unbalanced trees if geometry is clustered.
- **Object Median**: Splits the list of objects into two equal halves based on their centroids. Ensures balanced trees but may result in large overlapping bounding boxes.
- **Surface Area Heuristic (SAH)**: A greedy algorithm that attempts to minimize the expected traversal cost. It evaluates potential splits based on the probability of a ray hitting a child node (proportional to its surface area) and the cost of traversing it.

  $$Cost(b) = SA_L(b) \cdot N_L(b) + SA_R(b) \cdot N_R(b)$$

  where SA is the surface area and N is the number of primitives.

Comparison The choice of acceleration structure involves trade-offs between build time, memory usage, and traversal speed, as shown in Table 9.1.

9.4 Hardware-Accelerated Ray Tracing

Planned: BLAS/TLAS, ray query, RT cores.

9.5 Ray Differentials and Texture Filtering

Planned: mip-mapping, ray cones, needed for the lensed sky.

9.6 Environment and Sky Maps (opt.)

Planned: importance sampling, star catalogues.

Table 9.1: Comparison of Spatial Acceleration Structures[11]

Structure	**Memory**	**Build Time**	**Traversal Quality**
Regular Grid	Low – High	Low	Poor (non-uniform)
Quad/Octree	Low – High	Low	Good
K-d Tree	Low – High	Low – High	Good – Excellent
BVH	Low	Low – High	Good – Excellent

9.7 Real-Time GPU Design

Planned: compute vs. RT pipeline, progressive accumulation, temporal reuse, denoising (SVGF, OIDN).

10 Colour, Spectrum, Display

Scope: From photons to pixel values.

10.1 Human Vision and CIE 1931

Planned: XYZ, colour matching functions, metamerism.

10.2 Blackbody Radiation

Planned: Planck's law, Wien, Stefan–Boltzmann, Planckian locus.

10.3 RGB Colour Spaces

Planned: linear sRGB, transfer functions, Rec.709/2020/P3, gamut mapping.

10.4 Spectral Rendering

Planned: hero-wavelength sampling, RGB-to-spectrum upsampling, analytic CMFs.

10.5 Exposure and Adaptation

Planned: auto-exposure, white balance, Purkinje shift.

10.6 Tone Mapping

Planned: Reinhard, filmic, ACES, AgX, Khronos Neutral, LUT-based.

10.7 HDR Output (opt.)

Planned: PQ/HLG, BT.2100, reference white.

10.8 Post-Processing

Planned: bloom, glare PSF, lens flare, vignetting, grain, dithering and banding, motion blur.

Part IV

Physics

11 Units, Constants, Scales

Scope: Which numbers the simulation uses and why.

11.1 The Modern SI and Physical Constants

High-fidelity physical simulation engines require precise numerical representations of physical constants and quantities. These representations must accurately reflect the true values of the constants as established by international standards.

11.1.1 Defined and Experimental Constants

Following the 2019 revision of the International System of Units (SI) [3, 13], the definitions moved from artifact-based definitions to exact numerical definitions based on fundamental constants of nature.

Constants can be categorized into two types:

1. **Defined Constants:** These are constants whose numerical values are fixed by definition in the SI standard. They have zero measurement uncertainty. These include c, h, e, k_B, N_A, and others. These values are hardcoded as exact literals to the maximum precision supported by the data type. Many other constants, such as μ_0 and ϵ_0, are derived from these fixed constants and are also treated as exact values.
2. **Experimental Constants:** These constants are determined through empirical measurement and carry experimental uncertainties[12]. These include the gravitational constant (G), the fine-structure constant (α), and particle/object masses (e.g., electron mass m_e, hydrogen mass m_H, solar mass $M_\odot$). While CODATA provides recommended values along with their uncertainties, they are treated as floating-point values subject to experimental updates.

11.1.2 Simulation Engine Constants

To ensure reproducibility, determinism, and maximum numerical precision, the simulation engine initializes its internal physical constants using exact 2019 SI definitions[3, 13] and standard CODATA evaluated values[12] stored as 64-bit double-precision (or higher) floating-point numbers. The exact values can be referenced in the source code under `dependencies/gpp/src/core/units/units.cppm`.

11.2 Dimensional Analysis (opt.)

Planned: Buckingham π.

11.3 Geometrized Units

Planned: $GM/c^2 = 1.477\,\mathrm{km}$ and $GM/c^3 = 4.93\,\mu\mathrm{s}$ per $M_\odot$, Planck and natural units.

11.4 Astronomical Units

Planned: $M_\odot$, pc, ly, IAU nominal values.

11.5 Scale Table

Planned: stellar BH, Sgr A, M87*, horizon sizes, light-crossing times, typical disk temperatures.*

12 Special Relativity

Scope: Flat spacetime, the local limit of everything that follows.

12.1 Postulates and Lorentz Transformations

Planned: events, light cones, invariant interval.

12.2 Minkowski Space as a Flat Manifold

Planned: tensor chapter applied, proper time, world-lines.

12.3 Four-Velocity and Four-Momentum

Planned: photons: $p \cdot p = 0$, $E = h\nu$.

12.4 Dilation, Contraction, Rapidity

Planned: velocity addition, Thomas–Wigner rotation.

12.5 Doppler, Aberration, Beaming

Planned: transformation of I/ν^3 and of colour.

12.6 What You See vs. What Is

Planned: Terrell–Penrose rotation, light-travel time, simultaneity.

12.7 Accelerated Observers (opt.)

Planned: Rindler, Fermi–Walker transport.

12.8 Relativistic Dynamics

Planned: $T^{\mu\nu}$ of dust, fluid, EM field, $\partial_\mu T^{\mu\nu} = 0$.

13 Gravity as Geometry

Scope: General relativity in the amount needed here.

13.1 Newtonian Recap

Planned: potential, Kepler, tidal forces, escape velocity, Laplace/Michell dark stars, Newtonian deflection (half the GR value).

13.2 Equivalence Principle

Planned: weak/Einstein/strong, the elevator, local inertial frames.

13.3 Free Fall as Geodesic Motion

Planned: Newtonian limit $g_{tt} \approx -(1 + 2\Phi/c^2)$.

13.4 Einstein Field Equations

Planned: $G_{\mu\nu} = 8\pi G T_{\mu\nu}/c^4$, Λ, *vacuum, Hilbert action, energy conditions.*

13.5 Weak Field and Linearized Gravity

Planned: Shapiro delay, $4GM/(c^2 b)$ *(Eddington 1919), Lense–Thirring, gravitational waves.*

13.6 Gravitational Redshift and Clock Rates

Planned: static fields, $\sqrt{-g_{tt}}$.

13.7 Experimental Tests (mention)

Planned: Mercury's perihelion, Pound–Rebka, GPS, LIGO, EHT.

14 The Schwarzschild Black Hole

Scope: The simplest black hole, fully worked.

14.1 History

Planned: Schwarzschild 1916, Oppenheimer–Snyder, Wheeler.

14.2 The Solution

Planned: spherical symmetry, Birkhoff's theorem (statement), $r_s = 2GM/c^2$.

14.3 Coordinate Charts

Planned: Schwarzschild, isotropic, tortoise, Eddington–Finkelstein, Painlevé–Gullstrand, Kruskal–Szekeres, Penrose–Carter diagram.

14.4 Coordinate vs. Curvature Singularity

Planned: Kretschmann $48M^2/r^6$.

14.5 Event Horizon and Causal Structure

Planned: tilting light cones, trapped surfaces, infinite coordinate time vs. finite proper time.

14.6 Timelike Geodesics

Planned: E, L, effective potential, ISCO $6M$, marginally bound $4M$, precession, plunge, radial infall.

14.7 Null Geodesics

Planned: photon sphere $3M$, $b_c = 3\sqrt{3}M$, capture cross-section, Binet $u'' + u = 3Mu^2$, deflection integral, weak-field series, strong-field log divergence.

14.8 Shadow, Photon Ring, Einstein Ring

Planned: Lyapunov exponent, n-th image demagnification $e^{-\pi}$, relativistic images.

14.9 Redshift of Static and Orbiting Sources

Planned: $g(r)$.

14.10 Tides and Spaghettification

Planned: geodesic deviation, tidal radius, stellar vs. supermassive.

15 Charged and Rotating Black Holes

Scope: Reissner–Nordström, Kerr, Kerr–Newman.

15.1 Reissner–Nordström

Planned: two horizons, extremal, naked singularity, photon sphere.

15.2 Frame Dragging

Planned: Lense–Thirring effect.

15.3 Kerr in Boyer–Lindquist Coordinates

Planned: Σ, Δ, $a = J/M$, horizons, ergosphere, ring singularity, ZAMO, Penrose process, superradiance, area theorem.

15.4 Kerr–Schild Form

Planned: the ansatz, null congruence l_μ, Cartesian KS, implicit $r(x, y, z)$, algebraic inverse metric, regular across the horizon, sign conventions, spin-axis orientation.

15.5 Kerr–Newman and the No-Hair Theorems

Planned: uniqueness theorems.

15.6 Hidden Symmetry

Planned: Killing–Yano, Killing tensor, Carter constant $\mathcal{Q}$, separability of Hamilton–Jacobi.

15.7 Kerr Geodesics

Planned: Carter's equations, Mino time, $R(r)$ and $\Theta(\theta)$ potentials, turning points.

15.8 Equatorial Circular Orbits

Planned: Ω, E, L, ISCO formula (BPT72), efficiency η, spin limit 0.998.

15.9 Spherical Photon Orbits and the Kerr Shadow

Planned: Bardeen's (α, β), $\xi(r)$, $\eta(r)$, inclination dependence.

15.10 Extensions (opt.)

Planned: Kerr–(A)dS, extremal and super-extremal, "bumpy" non-Kerr metrics.

16 Light in Curved Spacetime

Scope: Relativistic optics for the camera.

16.1 Geometric-Optics Limit

Planned: eikonal, $k = \nabla S$ gives null geodesics, validity.

16.2 Null Geodesics as a Hamiltonian Flow

Planned: constants $E, L_z, \mathcal{Q}$, affine parameter.

16.3 Observers and Tetrads

Planned: static, ZAMO/LNRF, circular, infalling (Marck, rain frames), pixel to local direction to photon momentum, aberration of the whole sky.

16.4 Backward Tracing and Liouville

Planned: reversibility, deflection/lens map, I_ν/ν^3 invariant.

16.5 Frequency Shift

Planned: $g = (p \cdot u)_{\text{obs}}/(p \cdot u)_{\text{em}}$, special case $\sqrt{1 - 3M/r}$.

16.6 Ray Bundles and Image Distortion

Planned: geodesic deviation, Sachs optical scalars, magnification, caustics, ray differentials.

16.7 Light-Travel Time

Planned: Shapiro delay, image time delays, fast-light vs. slow-light.

16.8 Gravitational Lensing

Planned: thin-lens equation, Einstein radius, strong-field vs. strong-lens regime.

16.9 Polarization Transport (opt.)

Planned: Walker–Penrose constant, parallel transport.

16.10 Beyond Geometric Optics (mention)

Planned: wave lensing, diffraction.

17 Matter Around Black Holes

Scope: Orbits, accretion disks, particle models.

17.1 Test-Particle Motion

Planned: epicyclic frequencies, QPOs.

17.2 Accretion Basics

Planned: angular momentum, Eddington luminosity, efficiency η, Bondi.

17.3 Thin Disk Theory

Planned: Shakura–Sunyaev α, hydrostatic and energy balance, Newtonian $T \propto r^{-3/4}$.

17.4 Relativistic Thin Disk

Planned: Novikov–Thorne/Page–Thorne flux, zero-torque ISCO, T^4 peak at $\frac{49}{36} r_{\mathrm{in}}$, Luminet 1979.

17.5 Disk Appearance

Planned: multicolour blackbody, hardening factor ~ 1.7, X-ray vs. visible band mapping, limb darkening, returning radiation.

17.6 Thick and Hot Flows (mention)

Planned: slim disks, ADAF/RIAF, tori, jets (Blandford–Znajek), MRI, GRMHD (HARM, EHT library).

17.7 Particle Disk Models

Planned: pseudo-Newtonian (Paczyński–Wiita), geodesic particles, α-relaxation, Lynden-Bell–Pringle, emitters and sinks, SPH.

17.8 Companions and Tidal Disruption

Planned: Roche lobe (Eggleton), Hill sphere, mass transfer, Rees $t^{-5/3}$.

17.9 Stars Around Sgr A* and Black-Hole Growth (opt.)

Planned: S2 orbit, precession, spin evolution.

18 Relativistic Radiative Transfer

Scope: Emission, absorption, and what the camera records.

18.1 Photon Kinetic Theory

Planned: $f(x, p)$, Liouville equation, collision term.

18.2 Invariants

Planned: I/ν^3, j/ν^2, $\nu\alpha_\nu$, fluid-frame vs. observer-frame quantities.

18.3 Covariant Transfer Equation

Planned: optical depth, formal solution, exact constant-coefficient step update.

18.4 Emission Processes (opt.)

Planned: thermal/Kirchhoff, bremsstrahlung, synchrotron, Compton, Fe Kα.

18.5 Absorption and Scattering (opt.)

Planned: Thomson, Kramers.

18.6 From Specific Intensity to Image Flux

Planned: flux density, magnitudes.

18.7 Bridge to Rendering

Planned: volume rendering equation as the flat limit, g^4 RGB shortcut vs. spectral, $T_{\mathrm{obs}} = gT$, band mapping.

18.8 Polarized Transfer (opt.)

Planned: Stokes, Faraday rotation and conversion.

18.9 Numerical Considerations

Planned: $\tau \to 0$, optically thick limit.

19 Time and Clocks

Scope: Four different times in one simulation.

19.1 Four Times

Planned: coordinate, proper, observer/camera, simulation vs. wall clock.

19.2 Time Dilation Near Holes

Planned: u^t, the "frozen at the horizon" appearance.

19.3 Light-Travel Time

Planned: slow-light vs. fast-light, moving emitters, error budget.

19.4 Natural Time Scales

Planned: GM/c^3, ISCO period, light-crossing, viscous, thermal, tidal disruption, slow-motion and time-lapse design.

19.5 Time Stepping of Particles

Planned: coordinate-time vs. proper-time parametrization, determinism.

19.6 Shutter and Motion Blur (opt.)

Planned: exposure over t.

20 Many Bodies and Exotic Spacetimes

Scope: What has no closed form, and what is only for teaching.

20.1 The Two-Body Problem

Planned: no closed form, post-Newtonian, numerical relativity (BSSN, SXS), gravitational waves (LIGO).

20.2 Multi-Hole Approximations

Planned: Kerr–Schild superposition (flagged approximate), Majumdar–Papapetrou, Bohn et al. SXS lensing.

20.3 Exotic Metrics (opt.)

Planned: Ellis/Morris–Thorne wormholes, the Interstellar *wormhole, Alcubierre, Gödel, cosmic strings, boson stars, naked singularities, generic metric via AD.*

20.4 Cosmological Context (opt.)

Planned: FLRW, redshift z, Λ, Schwarzschild–de Sitter, primordial black holes.

21 Inside, Paradoxes, Universes, the Unknown

Scope: Established physics and speculation, clearly separated.

21.1 Falling In (mention)

Planned: nothing local at the horizon, r becomes timelike, spaghettification, the sky from inside (Hamilton), Doran frame.

21.2 Maximal Extension (mention)

Planned: Kruskal, white hole, Einstein–Rosen bridge (non-traversable), "other universes" as mathematical regions.

21.3 RN and Kerr Interiors (mention)

Planned: Cauchy horizon, mass inflation (Poisson–Israel), ring singularity, $r < 0$, closed timelike curves.

21.4 Singularities (mention)

Planned: Penrose–Hawking theorems, weak and strong cosmic censorship.

21.5 Paradoxes (mention)

Planned: outside vs. infalling observer, grandfather/Novikov consistency, chronology protection, information loss, firewalls, no-hair vs. entropy.

21.6 Thermodynamics and Quantum Effects (mention)

Planned: four laws, Bekenstein–Hawking $S = A/4$, Hawking temperature, evaporation time, Page curve, islands.

21.7 Universes (Speculative) (mention)

Planned: black-hole cosmology, baby universes (Smolin), Planck stars/bounces, multiverse.

21.8 Still Unknown (mention)

Planned: quantum gravity (strings/fuzzballs, LQG, asymptotic safety), singularity resolution, mass gap, SMBH seeds, tests of no-hair, dark matter and primordial BHs.

Part V

Rendering Curved Spacetime

22 Design Space and Solver Architecture

Scope: How the pieces of the previous parts become a renderer.

22.1 Pipeline Overview

Planned: camera tetrad, null ray, integrate, hit (disk/object/sky), g-shift, spectral, display.

22.2 Solver Families

Planned: Binet (exact Schwarzschild), Carter, Kerr–Schild Hamiltonian (default), generic AD, lens-map LUT, comparison table, decision rationale.

22.3 Kerr–Schild Hamiltonian Solver

Planned: ∂f, ∂l, RK4 vs. DP54, constraint handling.

22.4 Step Control and Termination

Planned: capture/escape/budget, far-field asymptotic remainder.

22.5 Lens-Map LUT Tier

Planned: parameterization $(a_*, r_{\mathrm{obs}}, \theta_{\mathrm{obs}})$, interpolation error, 3-D object limitation.

22.6 Scene Geometry

Planned: ray segments as chords, BVH/ray query, flat far field R_{flat} and its error bound.

22.7 Disks and Volumes Along Curved Rays

Planned: plane-crossing root finding, higher-order images, transfer integration.

22.8 Sky Lookup and Anti-Aliasing

Planned: ray-differential filtering, point stars.

22.9 Lighting in Curved Space

Planned: lights as g-shifted emitters, shading in the surface rest frame, what is dropped (GI).

22.10 Novel Contributions

Planned: to be defined together with the introduction.

23 Implementation

Scope: The MoleHole architecture.

23.1 Architecture

Planned: sim thread / render thread, `RenderSettings` *profiles (Flat/Lensed/Traced/Reference), pipeline stages.*

23.2 Precision

Planned: dimensionless f32, camera-relative, optional f64 reference, unit boundaries.

23.3 Shader Structure

Planned: equation-sheet headers.

23.4 Disk and Particle System

23.5 Spectral and Post-Processing Pipeline

23.6 Performance

Planned: per-pass timings, step counts, RTX 3060 numbers.

24 Validation and Results

Scope: Every physical claim traced to an analytic check.

24.1 Method

Planned: analytic references, convergence order, cross-code.

24.2 Analytic Test Suite

Planned: V-1… V-12: shadow, deflection, Bardeen curve, photon sphere, $E/L_z/Q$ drift, KS↔BL, $g(r)$, T^4 peak, step convergence, Einstein ring, pole crossing, GPU↔CPU parity.

24.3 Integrator Study

Planned: RK4 vs. DP54 vs. symplectic, error vs. cost.

24.4 Cross-Code and Observation

Planned: GYOTO, geokerr, DNGR figures, EHT ring diameters.

24.5 Performance and Scaling

24.6 Gallery

24.7 Limitations

Planned: "About this render" L2/L3 list.

25 Discussion, Future Work, Conclusion

25.1 Discussion

25.2 Future Work

25.3 Conclusion

Part VI

Appendices

A Notation and Conventions

Planned: signature, signs of a, index placement, Kerr–Schild l_μ signs.

Symbols

a Spin parameter $a = J/(Mc)$; dimensionless $a_* = a/M$.

c Speed of light in vacuum.

E, L_z, $\mathcal{Q}$ Conserved energy, axial angular momentum and Carter constant of a geodesic.

G Newtonian constant of gravitation.
$g_{\mu\nu}$ Spacetime metric, signature $(-+++)$.
g Frequency ratio $\nu_{\mathrm{obs}}/\nu_{\mathrm{em}}$ (redshift factor).
$\Gamma^{\mu}{}_{\alpha\beta}$ Christoffel symbols of the Levi-Civita connection.

I_ν Specific intensity; I_ν/ν^3 is invariant along rays.

λ Affine parameter along a geodesic.

M Mass of the black hole (length GM/c^2 in geometrized units).

Q Electric charge of the black hole.

Σ, Δ Kerr functions $\Sigma = r^2 + a^2\cos^2\theta$, $\Delta = r^2 - 2Mr + a^2 + Q^2$.

B Equation Sheet

Planned: Schwarzschild, Kerr, Kerr–Schild, Carter, ISCO, NT flux.

C Derivations

Planned: Schwarzschild Christoffels, Carter separation, ISCO, NT flux integral, T-peak 49/36.

D Constants Table

Planned: generated from `units.cppm`.

E Reproducibility

Planned: commands, presets, test IDs.

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Acronyms

AABB Axis-Aligned Bounding Box. 16

BRDF Bidirectional Reflectance Distribution Function. 13–15
BSDF Bidirectional Scattering Distribution Function. 13, 14
BTDF Bidirectional Transmittance Distribution Function. 13
BVH Bounding Volume Hierarchy. 16, 17

DNGR Double Negative Gravitational Renderer. 2, 3

EHT Event Horizon Telescope. 2, 3
EM Electro-Magnetic. 13

K-d Tree K-Dimensional Tree. 16, 17

OBB Oriented Bounding Box. 16

SAH Surface Area Heuristic. 16

VLBI Very Long Baseline Interferometry. 2